

UNIT-1LOGIC AND PROOFSProposition

logic-reasoning, argumentation.

Introduction:

We shall use only declarative sentences which have only one of the possible truth values called TRUE (T) or FALSE (F). Sometimes we denote them by 1 or 0.

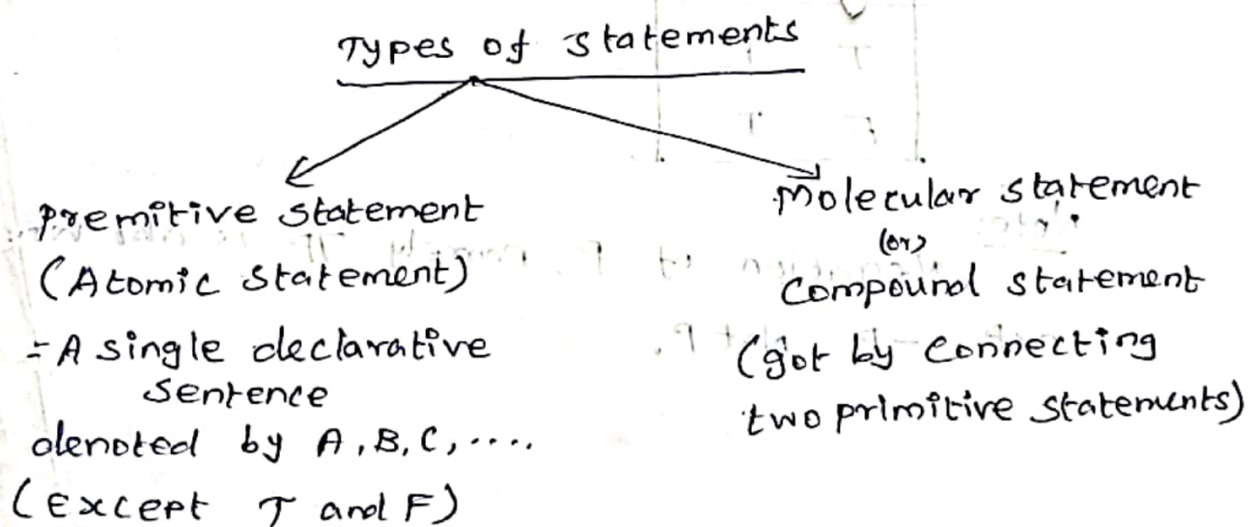
For Ex:

"India is a country" is a declarative sentence having a truth value T.

But "Do not speak" - is not a declarative sentence as it is a command. so it does not have a truth value.

Proposition (Statement)

Any declarative ~~ste~~ sentence is called a proposition (or) statement which is either true or false but not both.

Types of Statements:

Connectives:

Connective is an operation which is used to connect two or more than two statements. Simply it is called sentential connectives. It is also known as Logical Connectives or Logical operators.

Compound Statement:

Statements which contain one or more primary statements and some connectives are called compound or molecular or composite statements.

Basic Connectives

These are symbols used to connect two statements.

1. Negation (or) Denial

Symbol : $\neg$ (or) $\sim$

Meaning : Not

i.e. If 'P' is a statement, then $\neg P$ is "not P"

Ex: If P is "I am good"

Then $\neg P$ is "I am not good"

Table of Truth value

P	$\neg P$
T	F
F	T

Note:

Negation of P, namely $\neg P$ is also written as

$\sim P$, $-P$, Not P.

2. Disjunction:

Symbol : $\vee$

Meaning : 'OR'

Ex:

" $P \vee Q$ " means either P or Q .

Truth value

$P \vee Q$ is False only when both are False.

Truth Table:

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

Note:

Disjunction ' $\vee$ ' is a binary operator but Negation ' $\neg$ ' is a unary operator.

3. Conjunction:

Symbol : $\wedge$

Meaning : AND

Ex: " $P \wedge Q$ " means both P and Q .

Truth value:

$P \wedge Q$ is True only when both P and Q are True.

Truth table:

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Note:

If there are 'n' statements, then there are 2^n possibilities.

Ex: If $n=3$ statements, there are $2^3 = 8$ chances.

We shall now form the truth tables for different combinations of statements.

1. ^{construct} Truth table for $P \vee \neg Q$

P	Q	$\neg Q$	$P \vee \neg Q$
T	T	F	T
T	F	T	T
F	T	F	F
F	F	T	T

2. $P \wedge \neg P$

P	$\neg P$	$P \wedge \neg P$
T	F	F
F	T	F

3. $(P \vee Q) \vee \neg P$

P	Q	$\neg P$	$P \vee Q$	$(P \vee Q) \vee \neg P$
T	T	F	T	T
T	F	F	T	T
F	T	T	T	T
F	F	T	F	T

A. $\neg(\neg P \vee \neg Q)$

P	Q	$\neg P$	$\neg Q$	$\neg P \vee \neg Q$	$\neg(\neg P \vee \neg Q)$
T	T	F	F	F	T
T	F	F	T	T	F
F	T	T	F	T	F
F	F	T	T	T	F

Note: $\neg(\neg P \vee \neg Q) = P \wedge Q$ (check)

B. $\neg(\neg P \wedge \neg Q)$ [Note: Negation changes "AND" into "OR" and "OR" into "AND"]

P	Q	$\neg P$	$\neg Q$	$(\neg P \wedge \neg Q)$	$\neg(\neg P \wedge \neg Q)$
T	T	F	F	F	T
T	F	F	T	F	T
F	T	T	F	F	T
F	F	T	T	T	F

6.
 $\neg(\neg P) = P$
 $P \neg P \neg Q$
 $T \ F \ T$
 $F \ T \ F$

Note: $\neg(\neg P \wedge \neg Q) = P \vee Q$ (check)

1. Using the statements R : Mark is rich
 H : Mark is happy

Write the following in symbolic form.

(i) Mark is rich but unhappy
 $=$ Mark is rich and Mark is unhappy
 $= R \wedge \neg H$

(ii) Mark is poor but happy
 $= \neg R \wedge H$

(iii) Mark is rich or unhappy
 $= R \vee \neg H$

(iv) Mark is neither rich nor happy
 $=$ Mark is not rich and Mark is not happy
 $= \neg R \wedge \neg H$

v) Mark is rich or he is both poor and unhappy
 $= R \vee (\neg R \wedge \neg H)$

2. Form the truth tables

(i) $P \wedge (P \vee Q)$

P	Q	$P \vee Q$	$P \wedge (P \vee Q)$
T	T	T	T
T	F	T	T
F	T	T	F
F	F	F	F

(ii) $P \wedge (Q \wedge P)$

P	Q	$Q \wedge P$	$P \wedge (Q \wedge P)$
T	T	T	T
T	F	F	F
F	T	F	F
F	F	F	F

Further Connectives

④ Conditional statements:

If P and Q are two statements, then the statement $P \rightarrow Q$ which is read as "If P then Q " is called a "Conditional statements".

The statement $P \rightarrow Q$ has the truth value F , when Q has truth value F if P is True. Otherwise it has the truth value T .

Truth table:

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Here P is called "ANTECEDENT" and Q is called "CONSEQUENT"

Example:

Write the following in symbolic form

If either Jerry takes Calculus or Ken takes Sociology, then Larry will take English.

Let, J : Jerry takes Calculus
 K : Ken takes Sociology
 L : Larry will take English.

" $(J \vee K) \rightarrow L$ "

The crop will be destroyed if there is a flood

There is a flood: C

Crop will be destroyed: D

" $C \rightarrow D$ "

⑧ Form the truth table for $(P \rightarrow Q) \wedge (Q \rightarrow P)$

P	Q	$P \rightarrow Q$	$Q \rightarrow P$	$(P \rightarrow Q) \wedge (Q \rightarrow P)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

⑨ Bi-Conditional Statement:

If P and Q are two statements, then the statement $P \Leftrightarrow Q$, read as "P if and only if Q" or "P iff Q" is called a Biconditional statement.

The truth value of the biconditional statement $(P \Leftrightarrow Q)$ will be T whenever both P and Q have the same truth value.

Truth table:

P	Q	$P \Leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

Results:

Comparing with the truth table of $(P \rightarrow Q) \wedge (Q \rightarrow P)$ we see that it is the same as $(P \Leftrightarrow Q)$

$$\therefore (P \Leftrightarrow Q) \Leftrightarrow (P \rightarrow Q) \wedge (Q \rightarrow P)$$

where the symbol $\Leftrightarrow$ means "equivalent to"

Ex: Construct the truth table for the formula.

$$\text{① } \neg(P \wedge Q) \Leftrightarrow (\neg P \vee \neg Q)$$

P	Q	$\neg P$	$\neg Q$	$P \wedge Q$	$\neg(P \wedge Q)$	$\neg P \vee \neg Q$	$\neg(P \wedge Q) \Leftrightarrow (\neg P \vee \neg Q)$
T	T	F	F	T	F	F	T
T	F	F	T	F	T	T	T
F	T	T	F	F	T	T	T
F	F	T	T	F	T	T	T

$\therefore \neg(P \wedge Q) \Leftrightarrow (\neg P \vee \neg Q)$ is a Tautology.

Show that the truth values of the following formulas are independent of their components.

(i) $(P \wedge (P \rightarrow Q)) \rightarrow Q$

P	Q	$P \rightarrow Q$	$P \wedge (P \rightarrow Q)$	$(P \wedge (P \rightarrow Q)) \rightarrow Q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

(ii) $(P \rightarrow Q) \Leftrightarrow (\neg P \vee Q)$

P	Q	$\neg P$	$P \rightarrow Q$	$\neg P \vee Q$	$(P \rightarrow Q) \Leftrightarrow (\neg P \vee Q)$
T	T	F	T	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	T	T

(iii) $((P \rightarrow Q) \wedge (Q \rightarrow R)) \rightarrow (P \rightarrow R)$

P	Q	R	$P \rightarrow Q$	$Q \rightarrow R$	$(P \rightarrow Q) \wedge (Q \rightarrow R)$	$P \rightarrow R$	$((P \rightarrow Q) \wedge (Q \rightarrow R)) \rightarrow (P \rightarrow R)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	F	T	T
T	F	F	F	T	F	T	T
F	T	T	T	T	T	T	T
F	T	F	T	F	F	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

(iv) $(P \Leftrightarrow Q) \Leftrightarrow ((P \wedge Q) \vee (\neg P \wedge \neg Q))$

P	Q	$\neg P$	$\neg Q$	$P \Leftrightarrow Q$	$P \wedge Q$	$\neg P \wedge \neg Q$	$(P \wedge Q) \vee (\neg P \wedge \neg Q)$	$(P \Leftrightarrow Q) \Leftrightarrow ((P \wedge Q) \vee (\neg P \wedge \neg Q))$
T	T	F	F	T	T	F	T	T
T	F	F	T	F	F	F	F	T
F	T	T	F	F	F	F	F	T
F	F	T	T	T	F	T	T	T

Construct the truth table for the following:

$$\neg (Q \wedge (P \rightarrow Q)) \rightarrow P$$

P	Q	$P \rightarrow Q$	$Q \wedge (P \rightarrow Q)$	$\neg (Q \wedge (P \rightarrow Q)) \rightarrow P$
T	T	T	T	T
T	F	F	F	T
F	T	T	T	F
F	F	T	F	T

$$\neg (P \vee (Q \wedge R)) \Leftrightarrow ((P \vee R) \wedge (P \vee R))$$

1 P	2 Q	3 R	4 $Q \wedge R$	5 $P \vee (Q \wedge R)$	6 $\neg (P \vee (Q \wedge R))$	7 $P \vee R$	8 $P \vee R$	9 $(P \vee R) \wedge (P \vee R)$	10 Result
T	T	T	T	T	F	T	T	T	F
T	T	F	F	T	F	T	T	T	F
T	F	T	F	T	F	T	T	T	F
T	F	F	F	T	F	T	T	T	F
F	T	T	T	T	F	T	T	T	F
F	T	F	F	F	T	T	F	F	F
F	F	T	F	F	T	F	T	F	F
F	F	F	F	F	T	F	F	F	F

Given the truth value of P, Q as T and those of R, S as F. Find the truth values of the following.

$$(a) (\neg (P \wedge Q) \vee \neg R) \vee ((Q \rightarrow \neg P) \rightarrow (R \vee \neg S)) = T$$

$$(b) (P \rightarrow R) \wedge (\neg Q \rightarrow S) = F$$

$$(c) (P \vee (Q \rightarrow (R \wedge \neg P))) \Leftrightarrow (Q \vee \neg S) = T$$

Well formed formulas : (wff) :

Statement formula:

A statement formula is an expression which is a string having variables, parenthesis and connective symbols.

Rules for forming a wff:

⑥

- ①. A statement variable standing alone is a wff.
2. If A is a wff, then $\neg A$ is also a wff.
3. If A and B are two wffs, then $(A \vee B)$, $(A \wedge B)$, $(A \rightarrow B)$ and $(A \rightleftharpoons B)$ are wffs.
4. A string of symbols having the statement variables, connectives and Parenthesis is a wff, iff it can be got by applying rules 1, 2 and 3 a finite number of times.

Examples:

Not wff

Reason

1. $\neg P \wedge Q$

Absence of parenthesis $(\neg P \wedge Q)$ or $\neg(P \wedge Q)$

2. $(P \rightarrow Q) \rightarrow \neg \wedge Q$

$\wedge Q$ alone is not defined because $\wedge$ is a binary connective

3. $(P \rightarrow Q$

Absence of Right parenthesis $(P \rightarrow Q)$

4. $(P \wedge Q) \rightarrow Q$

Absence of left parenthesis $((P \wedge Q) \rightarrow Q)$

5. $(P \wedge Q) \rightarrow Q$

Absence of parenthesis $((P \wedge Q) \rightarrow Q)$

Note:

In practice, we omit the outer parenthesis.

For ex:

We write $P \wedge Q$ instead of $(P \wedge Q)$

We write $(P \wedge Q) \rightarrow Q$ instead of $((P \wedge Q) \rightarrow Q)$

Tautology:

A statement formula which is true regardless of the truth values of the statements in it is called a Tautology, (or) A logical truth.

Ex: $P \vee \neg P$ is a tautology.

Contradiction:

A statement formula which is False irrespective of the truth values of the statements in it is called a contradiction.

Ex: $P \wedge \neg P$ is a Contradiction.

P	$\neg P$	$P \wedge \neg P$
T	F	F
F	T	F

Note:

The negation of a contradiction is a tautology.

Here $P \wedge \neg P$ is F.

$\therefore \neg(P \wedge \neg P)$ is T.

Result:

The conjunction of two tautologies is a tautology

$$T \wedge T = T$$

From the formulas given below select which are tautology and which are contradiction. Using the definition and indicate again which one is a tautology or contradiction.

(a) $(P \rightarrow (P \vee Q))$ It is wdd

(b) $((P \rightarrow (\neg P)) \rightarrow \neg P)$ Not a wdd

(c) $((\neg Q \wedge P) \wedge Q)$ It is wdd

(d) $((P \rightarrow (Q \rightarrow R)) \rightarrow ((P \rightarrow Q) \rightarrow (P \rightarrow R)))$ It is wdd

(e) $((P \wedge Q) \rightarrow P)$ It is wdd

Contingency:

A statement formula which is neither tautology nor contradiction is called contingency.

Ex: $P \rightarrow Q$ is a Contingency.

Tautology	Contradiction	Fallacy	Contingency
In the result column all the entries are T	In the result column all the entries are F	In the result column any one entry is F	In the result column entries both are T and F
T	F	T	T
T	F	T	F
T	F	T	F
T	F	F	T

Equivalence of Formulas:

Def: If the truth values of two formulas A and B are the same in the final column, then we say that A and B are equivalent. This we write as $A \Leftrightarrow B$.

Ex:

- (i) $\neg\neg P$ is equivalent to P. $\therefore \neg\neg P \Leftrightarrow P$
 (ii) $P \vee \neg P$ is equivalent to T. $\therefore P \vee \neg P \Leftrightarrow T$
 (iii) $(P \wedge \neg P) \vee Q \Leftrightarrow Q$

P	Q	$\neg P$	$P \wedge \neg P$	$(P \wedge \neg P) \vee Q$
T	T	F	F	T
T	F	F	F	F
F	T	T	F	T
F	F	T	F	F

$\therefore (P \wedge \neg P) \vee Q \Leftrightarrow Q$

(iv) $(P \vee \neg P) \Leftrightarrow (Q \vee \neg Q)$

P	Q	$\neg P$	$\neg Q$	$P \vee \neg P$	$Q \vee \neg Q$
T	T	F	F	T	T
T	F	F	T	T	T
F	T	T	F	T	T
F	F	T	T	T	T

$\therefore (P \vee \neg P) \Leftrightarrow (Q \vee \neg Q)$

Result:

To show that $A \Leftrightarrow B$, it is enough to show that $A \Rightarrow B$ is a Tautology.

$\therefore A \Leftrightarrow B$ iff $A \Rightarrow B$ is Tautology.

Ex:

⊗ Show that $(P \rightarrow Q) \Leftrightarrow (\neg P \vee Q)$

P	Q	$\neg P$	$P \rightarrow Q$	$\neg P \vee Q$	$(P \rightarrow Q) \Leftrightarrow (\neg P \vee Q)$
T	T	F	T	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	T	T

As $(P \rightarrow Q) \Leftrightarrow (\neg P \vee Q)$ is a Tautology

$\therefore (P \rightarrow Q) \Leftrightarrow (\neg P \vee Q)$

We shall now give below the ^{list} ~~result~~ of basic equivalent formulas which are useful. Here T and F are used in the sense that T can be replaced only by ~~the~~ a Tautology and F can be replaced only by a contradiction.

(1) $P \vee P \Leftrightarrow P$
 $P \wedge P \Leftrightarrow P$ } Idempotent Laws

(2) $(P \vee Q) \vee R \Leftrightarrow P \vee (Q \vee R)$
 $(P \wedge Q) \wedge R \Leftrightarrow P \wedge (Q \wedge R)$ } Associative Laws

(3) $P \vee Q \Leftrightarrow Q \vee P$
 $P \wedge Q \Leftrightarrow Q \wedge P$ } Commutative Laws

(4) $P \wedge (Q \vee R) \Leftrightarrow (P \wedge Q) \vee (P \wedge R)$
 $P \vee (Q \wedge R) \Leftrightarrow (P \vee Q) \wedge (P \vee R)$ } Distributive Laws

$$\left. \begin{aligned} (5) \quad & P \vee F \Leftrightarrow P \\ & P \wedge T \Leftrightarrow P \end{aligned} \right\} \text{Identity Laws}$$

$$\left. \begin{aligned} (6) \quad & P \vee T \Leftrightarrow T \\ & P \wedge F \Leftrightarrow F \end{aligned} \right\} \text{Dominance Laws}$$

$$\left. \begin{aligned} (7) \quad & P \vee \neg P \Leftrightarrow T \\ & P \wedge \neg P \Leftrightarrow F \end{aligned} \right\} \text{Complement Laws}$$

$$\left. \begin{aligned} (8) \quad & P \vee (P \wedge Q) \Leftrightarrow P \\ & P \wedge (P \vee Q) \Leftrightarrow P \end{aligned} \right\} \text{Absorption Laws}$$

$$\left. \begin{aligned} (9) \quad & \neg(P \vee Q) \Leftrightarrow \neg P \wedge \neg Q \\ & \neg(P \wedge Q) \Leftrightarrow \neg P \vee \neg Q \end{aligned} \right\} \text{De Morgan's Laws}$$

Note:

By the definition of equivalence, we can replace one formula by its equivalent formula.

For ex:

Consider the formula $P \wedge Q$

$$\text{But } P \Leftrightarrow P \vee P$$

$$\therefore P \wedge Q \Leftrightarrow (P \vee P) \wedge Q$$

Example:

$$(X) \text{ 1. Show that } P \rightarrow (Q \rightarrow R) \Leftrightarrow P \rightarrow (\neg Q \vee R) \Leftrightarrow (P \wedge Q) \rightarrow R$$

$$\text{Proof: } P \rightarrow (Q \rightarrow R)$$

$$\Leftrightarrow P \rightarrow (\neg Q \vee R) \xrightarrow{(1)} \because P \rightarrow Q \Leftrightarrow \neg P \vee Q \quad \text{Conditional as disjunction}$$

$$\Leftrightarrow \neg P \vee (\neg Q \vee R)$$

$$\Leftrightarrow (\neg P \vee \neg Q) \vee R \quad \because \text{Associative Law}$$

$$\Leftrightarrow \neg(P \wedge Q) \vee R \quad \because \text{De Morgan's Law}$$

$$\Leftrightarrow (P \wedge Q) \rightarrow R \xrightarrow{(2)} \because P \rightarrow Q \Leftrightarrow \neg P \vee Q$$

From (1) and (2) we get,

$$P \rightarrow (Q \rightarrow R) \Leftrightarrow P \rightarrow (\neg Q \vee R) \Leftrightarrow (P \wedge Q) \rightarrow R$$

2. Show that $(\neg P \wedge (\neg Q \wedge R)) \vee (Q \wedge R) \vee (P \wedge R) \Leftrightarrow R$

Solution:

$$(\neg P \wedge (\neg Q \wedge R)) \vee (Q \wedge R) \vee (P \wedge R)$$

$$\Leftrightarrow (\neg P \wedge (\neg Q \wedge R)) \vee (Q \vee P) \wedge R \text{ by Distributive Law}$$

$$\Leftrightarrow (\neg P \wedge (\neg Q \wedge R)) \vee (P \vee Q) \wedge R \text{ by Commutative Law}$$

$$\Leftrightarrow ((\neg P \wedge \neg Q) \wedge R) \vee (P \vee Q) \wedge R \text{ by Associative Law}$$

$$\Leftrightarrow (\neg(P \vee Q) \wedge R) \vee (P \vee Q) \wedge R \text{ by De Morgan's Law}$$

$$\Leftrightarrow (\neg(P \vee Q) \vee (P \vee Q)) \wedge R \text{ by Distributive Law}$$

$$\Leftrightarrow ((P \vee Q) \vee \neg(P \vee Q)) \wedge R \text{ by Commutative Law}$$

$$\Leftrightarrow T \wedge R$$

$$\therefore P \vee \neg P \Leftrightarrow T$$

$$T \wedge P \Leftrightarrow P$$

$$\Leftrightarrow R$$

Thus

$$(\neg P \wedge (\neg Q \wedge R)) \vee (Q \wedge R) \vee (P \wedge R) \Leftrightarrow R$$

3. Show that $((P \vee Q) \wedge \neg(\neg P \wedge (\neg Q \vee \neg R))) \vee (\neg P \wedge \neg Q) \vee (\neg P \wedge \neg R)$ is a Tautology.

Sol:

$$((P \vee Q) \wedge \neg(\neg P \wedge (\neg Q \vee \neg R))) \vee (\neg P \wedge \neg Q) \vee (\neg P \wedge \neg R)$$

$$\Leftrightarrow ((P \vee Q) \wedge \neg(\neg P \wedge (\neg Q \vee \neg R))) \vee (\neg(P \vee Q) \vee \neg(P \vee R))$$

by De Morgan's Law

$$\Leftrightarrow ((P \vee Q) \wedge \neg(\neg P \wedge (\neg Q \vee \neg R))) \vee \neg((P \vee Q) \wedge (P \vee R))$$

by De Morgan's Law

$$\Leftrightarrow ((P \vee Q) \wedge \neg(\neg P \wedge (\neg Q \vee \neg R))) \vee \neg(P \vee (Q \wedge R))$$

by Distributive Law

$$\Leftrightarrow ((P \vee Q) \wedge \neg(\neg P \wedge \neg(Q \wedge R))) \vee \neg(P \vee (Q \wedge R))$$

by De Morgan's Law

$$\Leftrightarrow ((P \vee Q) \wedge (P \vee (Q \wedge R))) \vee \neg(P \vee (Q \wedge R)) \text{ by } \neg P \vee P \Leftrightarrow T$$

$$\Leftrightarrow ((P \vee Q) \wedge (P \vee Q) \wedge (P \vee R)) \vee \neg(P \vee (Q \wedge R))$$

by Distributive Law

$$\Leftrightarrow T$$

$$\begin{aligned} &\Leftrightarrow ((P \vee Q) \wedge (P \vee R)) \vee \neg(P \vee (Q \wedge R)) \text{ by Idempotent Law} \\ &\Leftrightarrow (P \vee (Q \wedge R)) \vee \neg(P \vee (Q \wedge R)) \text{ by Distributive Law} \\ &\Leftrightarrow T \quad \therefore (P \vee \neg P) \Leftrightarrow T \end{aligned}$$

Note:

We can use the same result by using truth table.

4. Prove that $(P \rightarrow Q) \wedge (R \rightarrow Q) \Leftrightarrow (P \vee R) \rightarrow Q$

Sol:

Now, $(P \rightarrow Q) \wedge (R \rightarrow Q)$

$$\Leftrightarrow (\neg P \vee Q) \wedge (\neg R \vee Q) \quad \because P \rightarrow Q \Leftrightarrow \neg P \vee Q$$

$$\Leftrightarrow (\neg P \wedge \neg R) \vee Q \quad \text{by Distributive Law}$$

$$\Leftrightarrow \neg(P \vee R) \vee Q \quad \text{by DeMorgan's Law}$$

$$\Leftrightarrow (P \vee R) \rightarrow Q \quad \because P \rightarrow Q \Leftrightarrow \neg P \vee Q$$

5. Show that $P \rightarrow (Q \vee R) \Leftrightarrow (P \rightarrow Q) \vee (P \rightarrow R)$

Sol:

Now, $(P \rightarrow Q) \vee (P \rightarrow R)$

$$\Leftrightarrow (\neg P \vee Q) \vee (\neg P \vee R) \quad \because P \rightarrow Q \Leftrightarrow \neg P \vee Q$$

$$\Leftrightarrow \neg P \vee (Q \vee R) \quad \text{by Distributive Law}$$

$$\Leftrightarrow P \rightarrow (Q \vee R) \quad \because P \rightarrow Q \Leftrightarrow \neg P \vee Q$$

6. Show that $P \rightarrow (Q \rightarrow P) \Leftrightarrow \neg P \rightarrow (P \rightarrow Q)$

Sol:

Now, $\neg P \rightarrow (P \rightarrow Q)$

and $P \rightarrow (Q \rightarrow P)$

$$\Leftrightarrow \neg P \rightarrow (\neg P \vee Q)$$

$$\Leftrightarrow P \rightarrow (\neg Q \vee P)$$

$$\Leftrightarrow P \vee (\neg P \vee Q)$$

$$\Leftrightarrow \neg P \vee (\neg Q \vee P)$$

$$\Leftrightarrow (P \vee \neg P) \vee Q$$

$$\Leftrightarrow \neg P \vee (P \vee \neg Q)$$

$$\Leftrightarrow T \vee Q$$

$$\Leftrightarrow (\neg P \vee P) \vee \neg Q$$

$$\Leftrightarrow (T \rightarrow \textcircled{1})$$

$$\Leftrightarrow T \vee \neg Q$$

$$\Leftrightarrow T \rightarrow \textcircled{2}$$

$\therefore$ From $\textcircled{1}$ and $\textcircled{2}$ we have

$$P \rightarrow (Q \rightarrow P) \Leftrightarrow \neg P \rightarrow (P \rightarrow Q)$$

Tautological Implications:

Def:

Given $P \rightarrow Q$, the statement

- (i) $Q \rightarrow P$ is called its Converse
- (ii) $\neg P \rightarrow \neg Q$ is called its Inverse
- (iii) $\neg Q \rightarrow \neg P$ is called its Contrapositive

Ex:

"If Dhoni is Captain, then Team India wins"

P : Dhoni is captain

Q : Team India wins

Converse:

"If Team India wins, then Dhoni is Captain"

$$\text{is } Q \rightarrow P.$$

Inverse: Contrapositive

If Team India does not win, then Dhoni is not Captain

$$\text{is } \neg Q \rightarrow \neg P$$

Inverse:

If Dhoni is not Captain, then Team India does not win

$$\text{is } \neg P \rightarrow \neg Q.$$

Def: (Tautological Implications)

A statement 'A' is said to be Tautologically

imply a statement 'B' iff $A \rightarrow B$ is a Tautology.

We shall denote this by $A \Rightarrow B$ as read 'A implies B'.

Now, consider the table

P	Q	$\neg P$	$\neg Q$	$(P \rightarrow Q)$	$Q \rightarrow P$	$\neg P \rightarrow \neg Q$	$\neg Q \rightarrow \neg P$
T	T	F	F	T	T	T	T
T	F	F	T	F	F	T	F
F	T	T	F	T	T	F	T
F	F	T	T	T	T	T	T

Note that $(P \rightarrow Q) \Leftrightarrow (\neg Q \rightarrow \neg P)$

$(Q \rightarrow P) \Leftrightarrow (\neg P \rightarrow \neg Q)$

Note

(i) Just as $A \Leftrightarrow B$ means $A \Rightarrow B$ is a Tautology,

$A \Rightarrow B$ means that $A \rightarrow B$ is a Tautology.

(ii) Just as $\Leftrightarrow$ is not a connective, $\Rightarrow$ is also not a connective.

Alternative definition of Equivalence

If $A \Rightarrow B$ and $B \Rightarrow A$, then A and B are equivalent.

Ex: Prove the following implication

$$(i) (P \wedge Q) \Rightarrow (P \rightarrow Q)$$

Sol: To prove this we have to show that

$(P \wedge Q) \Rightarrow (P \rightarrow Q)$ is a Tautology.

P	Q	$P \wedge Q$	$P \rightarrow Q$	$(P \wedge Q) \Rightarrow (P \rightarrow Q)$
---	---	--------------	-------------------	--

T	T	T	T	T
---	---	---	---	---

T	F	F	F	T
---	---	---	---	---

F	T	F	T	T
---	---	---	---	---

F	F	F	T	T
---	---	---	---	---

$$\therefore (P \wedge Q) \Rightarrow (P \rightarrow Q)$$

$$(ii) P \Rightarrow (Q \rightarrow P)$$

Sol: To prove this we have to show that $P \Rightarrow (Q \rightarrow P)$ is a Tautology.

P	Q	$Q \rightarrow P$	$P \Rightarrow (Q \rightarrow P)$
---	---	-------------------	-----------------------------------

T	T	T	T
---	---	---	---

T	F	T	T
---	---	---	---

F	T	F	T
---	---	---	---

F	F	T	T
---	---	---	---

$$\therefore P \Rightarrow (Q \rightarrow P)$$

$$(ii) (P \rightarrow (Q \rightarrow R)) \Rightarrow ((P \rightarrow Q) \rightarrow (P \rightarrow R))$$

Sol:

P	Q	R	$Q \rightarrow R$	$P \rightarrow (Q \rightarrow R)$	$P \rightarrow Q$	$P \rightarrow R$	$(P \rightarrow Q) \rightarrow (P \rightarrow R)$
T	T	T	T	T	T	T	T
T	T	F	F	F	F	F	T
T	F	T	T	T	F	T	T
T	F	F	T	T	F	F	T
F	T	T	T	T	T	T	T
F	T	F	F	T	T	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

As $(P \rightarrow (Q \rightarrow R)) \rightarrow ((P \rightarrow Q) \rightarrow (P \rightarrow R))$ is a Tautology
 $\therefore (P \rightarrow (Q \rightarrow R)) \Rightarrow ((P \rightarrow Q) \rightarrow (P \rightarrow R))$

Duality Law:

Def: Two formulas A and A* are said to be duals of each other, if one is got from the other by replacing $\wedge$ by $\vee$, $\vee$ by $\wedge$, T by F and F by T.

Ex:

Formula A	Dual A*
(a) $(P \vee Q) \wedge R$	$(P \wedge Q) \vee R$
(b) $(P \wedge Q) \vee T$	$(P \vee Q) \wedge F$
(c) $\neg(P \vee Q) \wedge (P \vee \neg(Q \wedge T))$	$\neg(P \wedge Q) \vee (P \wedge \neg(Q \vee T))$

Result:

If A* and B* are Duals of A and B, then
 $A \Leftrightarrow B$ means $A^* \Leftrightarrow B^*$

1. Show that $\neg(P \wedge Q) \rightarrow (\neg P \vee (\neg P \vee Q)) \Leftrightarrow (\neg P \vee Q)$

Sol:

$$\neg(P \wedge Q) \rightarrow (\neg P \vee (\neg P \vee Q))$$

$$\Leftrightarrow \neg(P \wedge Q) \rightarrow ((\neg P \vee \neg P) \vee Q) \text{ by Associative Law}$$

$$\Leftrightarrow \neg(P \wedge Q) \rightarrow (\neg P \vee Q)$$

$$\because P \vee P \Leftrightarrow P$$

$$\Leftrightarrow \neg(\neg(P \wedge Q)) \vee (\neg P \vee Q)$$

$$\because P \rightarrow Q \Leftrightarrow \neg P \vee Q$$

$$\Leftrightarrow (P \wedge Q) \vee (\neg P \vee Q)$$

$$\because \neg \neg P \Leftrightarrow P$$

$$\Leftrightarrow ((P \wedge Q) \vee \neg P) \vee Q$$

by Associative Law

$$\Leftrightarrow ((P \vee \neg P) \wedge (Q \vee \neg P)) \vee Q$$

by Distributive Law

$$\Leftrightarrow (T \wedge (Q \vee \neg P)) \vee Q$$

$$P \vee \neg P \Leftrightarrow T$$

$$\Leftrightarrow (Q \vee \neg P) \vee Q$$

$$T \wedge P \Leftrightarrow P$$

$$\Leftrightarrow (\neg P \vee Q) \vee Q$$

by Commutative Law

$$\Leftrightarrow \neg P \vee (Q \vee Q)$$

by Associative Law

$$\Leftrightarrow (\neg P \vee Q)$$

$$\because Q \vee Q \Leftrightarrow Q \text{ Idempotent Law}$$

$\equiv$

2. Prove that $(P \vee Q) \wedge (\neg P \wedge (\neg P \wedge Q)) \Leftrightarrow (\neg P \wedge Q)$

Other Connectives

We have already defined the connectives $\neg, \wedge, \vee, \rightarrow, \Leftrightarrow$

Now we introduce some other connectives. There are two important connectives namely NAND and NOR.

NAND:

The word NAND is a combination of 'NOT' and 'AND'.

The connective 'NAND' is denoted by the symbol $\uparrow$.

For any two formulae P and Q, $P \uparrow Q$ is defined as

$$P \uparrow Q = \neg(P \wedge Q)$$

Truth Table

P	Q	$P \uparrow Q$
T	T	F
T	F	T
F	T	T
F	F	T

NOR:

The word 'NOR' is a combination of 'NOT' and 'OR'.
The Connective 'NOR' is denoted by the symbol $\downarrow$.
For any two formulae P and Q , $P \downarrow Q$ is defined as
$$P \downarrow Q = \neg(P \vee Q)$$

Truth table

P	Q	$P \downarrow Q$
T	T	F
T	F	F
F	T	F
F	F	T

NORMAL FORMS:

In this section we will call the Connective 'AND' $\wedge$ by Product and the Connective 'OR' $\vee$ by Sum.

Given two statements, P and Q , then $P \wedge \neg P$, $P \wedge Q$, $Q \wedge \neg Q$ etc are called elementary Products.

Their sum, for example

$(P \wedge \neg P) \vee (P \wedge Q) \vee (Q \wedge \neg Q)$ is the sum of elementary Products.

It is called ~~disjunctive~~ Disjunctive Normal Form (DNF)

Given two statements P and Q , then $P \vee \neg P$, $P \vee Q$,

$Q \vee \neg Q$, etc are called elementary Sums.

Their product, for example

$(P \vee \neg P) \wedge (P \vee Q) \wedge (Q \vee \neg Q)$ is the Product of elementary Sums

It is called the Conjunctive Normal Form (CNF)

Given a formula, changing it into any one of the above two normal forms is called Reduction to Normal Forms

Definition (Disjunctive Normal Form)

A formula is said to be in disjunctive normal form if it is,

- Equivalent to a given formula
- It contains sum of elementary products.

Example:

1. Find the disjunctive normal form of the following

1a) $P \wedge (P \rightarrow Q)$

Sol:

$$P \wedge (P \rightarrow Q)$$

$$\Leftrightarrow P \wedge (\neg P \vee Q) \quad \text{Distributive Law}$$

$$\Leftrightarrow (P \wedge \neg P) \vee (P \wedge Q)$$

Which is the sum of products.

$$P \wedge (P \rightarrow Q) \Leftrightarrow (P \wedge \neg P) \vee (P \wedge Q)$$

This is disjunctive normal form.

(b) $(\neg(P \vee Q)) \Leftrightarrow (P \wedge Q)$

Sol:

We use the formula

$$(R \Leftrightarrow S) \Leftrightarrow (R \wedge S) \vee (\neg R \wedge \neg S)$$

$\therefore$ The above becomes, Take $R = \neg(P \vee Q)$ and $S = (P \wedge Q)$

$$(\neg(P \vee Q)) \Leftrightarrow (\neg(P \vee Q) \wedge (P \wedge Q)) \vee ((\neg(P \vee Q)) \wedge \neg(P \wedge Q))$$

$$\Leftrightarrow (\neg(P \vee Q) \wedge (P \wedge Q)) \vee ((\neg(P \vee Q)) \wedge \neg(P \wedge Q))$$

$$\Leftrightarrow ((\neg P \wedge \neg Q) \wedge (P \wedge Q)) \vee ((\neg P \vee \neg Q) \wedge (\neg P \vee \neg Q))$$

by De Morgan's Law

$$\Leftrightarrow (\neg P \wedge \neg Q \wedge P \wedge Q) \vee ((\neg P \vee \neg Q) \wedge (\neg P \vee \neg Q))$$

$$\Leftrightarrow (\neg P \vee \neg Q) \wedge (\neg P \vee \neg Q) \wedge (\neg P \vee \neg Q) \wedge (\neg P \vee \neg Q)$$

by distributive Law

$$\Leftrightarrow (\neg P \wedge \neg Q \wedge P \wedge Q) \vee (\neg P \wedge \neg P) \vee (\neg Q \wedge \neg P) \vee (\neg P \wedge \neg Q) \vee (\neg Q \wedge \neg Q)$$

$\therefore$ This is sum of products.

Conjunctive Normal Form : (Product of sums)

If a formula is,

(i) equivalent to a given formula

(ii) The product of elementary sums.

Then the given formula is in Conjunctive Normal Form (CNF).

1. Write the following in Conjunctive normal form.

(a) $P \wedge (P \rightarrow Q)$

Sol:

$$P \wedge (P \rightarrow Q)$$

$$\Leftrightarrow (P \wedge (\neg P \vee Q))$$

$$\Leftrightarrow (P \vee P) \wedge (\neg P \vee Q) \quad \because P \vee P \Leftrightarrow P$$

This is the product of sums.

$\Rightarrow$ This is a conjunctive normal form.

(b) $(\neg(P \vee Q) \Leftrightarrow (P \wedge Q))$

Sol: We use the formula,

$$(R \Leftrightarrow S) \Leftrightarrow (R \rightarrow S) \wedge (S \rightarrow R)$$

Take $R = \neg(P \vee Q)$ and $S = (P \wedge Q)$

The above becomes,

$$(\neg(P \vee Q) \Leftrightarrow (P \wedge Q)) \Leftrightarrow (\neg(P \vee Q) \rightarrow (P \wedge Q)) \wedge ((P \wedge Q) \rightarrow \neg(P \vee Q))$$

$$\Leftrightarrow ((\neg(P \vee Q) \vee (P \wedge Q)) \wedge (\neg(P \wedge Q) \vee \neg(P \vee Q))$$

$$\Leftrightarrow (P \vee Q \vee P) \wedge (P \vee Q \vee Q) \wedge ((\neg P \vee \neg Q) \vee (\neg P \wedge \neg Q))$$

Demorgan's Law

$$\Leftrightarrow (P \vee Q \vee P) \wedge (P \vee Q \vee Q) \wedge (\neg P \vee \neg Q \vee \neg P) \wedge$$

$$(\neg P \vee \neg Q \vee \neg Q)$$

by distributive Law

This is the product of sums and so is

conjunctive normal form.

Definition: (Disjunctive Normal Form)

Principle Disjunctive Normal Forms: (PDNF)

i) Given two statements P and Q. we form all possible formulas which contain conjunction of P with either Q (or) $\neg Q$ (or) Conjunction of Q with either P (or) $\neg P$.

ii only $P \wedge Q, P \wedge \neg Q, \neg P \wedge Q, \neg P \wedge \neg Q$ are allowed

But $P \wedge P, Q \wedge \neg Q$ etc are not allowed.

We called $P \wedge Q, P \wedge \neg Q, \neg P \wedge Q, \neg P \wedge \neg Q$ as ^(minimum) "MIN TERMS"

iii If a min term appears twice after commutative law, we drop one term.

iv if we get $P \wedge Q, Q \wedge P$ we drop one term.

Definition: [Principle Disjunctive Normal Form]:

For a given formula an equivalent formula having disjunctions of MIN TERMS only. is called it

"Principle Disjunctive Normal Form". It is also called known as the "Sum of Products Canonical form".

In a similar manner given two statements, P and Q we form the sums $P \vee Q, P \vee \neg Q, \neg P \vee Q$ and $\neg P \vee \neg Q$. These are called MAX TERM.

As above $P \vee P, P \vee \neg P, Q \vee Q$ are not allowed and two similar terms are taken with one.

Principal Conjunctive Normal Form: (PCNF)

If a formula is equivalent another formula having conjunctions of max-terms only. It is known as the principal conjunctive Normal Form. (Or) the Product of sums canonical form.

To write the Principal disjunctive normal form:

Given P and Q , the minterms are $P \wedge Q, P \wedge \neg Q, \neg P \wedge Q, \neg P \wedge \neg Q$, the truth values of these four minterms are as follows:

P	Q	$P \wedge Q$	$P \wedge \neg Q$	$\neg P \wedge Q$	$\neg P \wedge \neg Q$
T	T	T	F	F	F
T	F	F	T	F	F
F	T	F	F	T	F
F	F	F	F	F	T

Note that, each of the minterm has the value T in only one place.

So, given a formula write the truth table for it.

In the final column, replace T in a position by the corresponding minterm which has T in that position.

Take the disjunction of these minterms.

It gives the principal Disjunctive Normal Form.

Ex:

Write the PDNF for the following formula

$$P \rightarrow Q, P \vee Q, \neg(P \wedge Q)$$

Truth Table is,

P	Q	$P \rightarrow Q$	$P \vee Q$	$P \wedge Q$	$\neg(P \wedge Q)$
T	T	T	T	T	F
T	F	F	T	F	T
F	T	T	T	F	T
F	F	T	F	F	T

∴ The PDNF's are

$$(i) P \rightarrow Q \Leftrightarrow (P \wedge Q) \vee (\neg P \wedge Q) \vee (\neg P \wedge \neg Q)$$

$$(ii) P \vee Q \Leftrightarrow (P \wedge Q) \vee (P \wedge \neg Q) \vee (\neg P \wedge Q)$$

$$(iii) \neg(P \wedge Q) \Leftrightarrow (P \wedge \neg Q) \vee (\neg P \wedge Q) \vee (\neg P \wedge \neg Q)$$

We want to write the PDNF of a given formula without using the truth table, we follow the method given below:

Step-1 Replace the conditional ($\rightarrow$) and biconditional ($\Leftrightarrow$) by $\vee, \wedge, \neg$.

Step-2 Drop any elementary product which is a contradiction (eg. $P \wedge \neg P \Leftrightarrow F$).

Step-3 Introduce missing factors in the disjunction.

Step-4 If two factors are identical, drop one term.

$$\neg(P \wedge Q) \rightarrow (\neg P \vee (\neg P \vee \neg Q))$$

Ex:

1. Find the PDNF of $\neg P \vee Q$

Sol:

$$\neg P \vee Q$$

$$\Leftrightarrow (\neg P \wedge T) \vee (Q \wedge T) \quad \text{Introduce missing Term}$$

$$\Leftrightarrow (\neg P \wedge (Q \vee \neg Q)) \vee (Q \wedge (P \vee \neg P)) \quad P \vee \neg P \Leftrightarrow T, Q \vee \neg Q \Leftrightarrow T.$$

$$\Leftrightarrow (\neg P \wedge Q) \vee (\neg P \wedge \neg Q) \vee (Q \wedge P) \vee (Q \wedge \neg P) \quad \text{distributive Law}$$

$$\Leftrightarrow (\neg P \wedge Q) \vee (\neg P \wedge \neg Q) \vee (P \wedge Q) \vee (\neg P \wedge Q) \quad \text{Commutative Law.}$$

$$\Leftrightarrow (\neg P \wedge Q) \vee (\neg P \wedge \neg Q) \vee (P \wedge Q)$$

2. Obtain the PDNF of $P \rightarrow ((P \rightarrow Q) \wedge \neg(\neg Q \vee \neg P))$

Sol: Use the formula " $P \rightarrow Q \Leftrightarrow \neg P \vee Q$ "

$$P \rightarrow ((P \rightarrow Q) \wedge \neg(\neg Q \vee \neg P))$$

$$\Leftrightarrow P \rightarrow ((\neg P \vee Q) \wedge (Q \wedge P)) \quad \because \neg \neg P \Leftrightarrow P, \neg \neg Q \Leftrightarrow Q$$

$$\Leftrightarrow \neg P \vee ((\neg P \vee Q) \wedge (Q \wedge P))$$

$$\Leftrightarrow \neg P \vee ((\neg P \wedge Q \wedge P) \vee (Q \wedge Q \wedge P)) \quad \text{Distributive Law}$$

$$\Leftrightarrow \neg P \vee ((\neg P \wedge P \wedge Q) \vee (Q \wedge P)) \quad \text{Commutative Law}$$

$$\Leftrightarrow \neg P \vee ((F \wedge Q) \vee (P \wedge Q))$$

$$\Leftrightarrow \neg P \vee (P \wedge Q) \quad \text{as } (F \wedge Q) \Leftrightarrow F \text{ and is dropped}$$

Introduce missing Term

$$\Leftrightarrow (\neg P \wedge T) \vee (P \wedge Q)$$

$$\Leftrightarrow (\neg P \wedge (Q \vee \neg Q)) \vee (P \wedge Q)$$

$$\Leftrightarrow (\neg P \wedge Q) \vee (\neg P \wedge \neg Q) \vee (P \wedge Q) \quad \text{Distributive Law.}$$

==
This is the PDNF.

3. Obtain the PDNF of $(P \wedge Q) \vee (\neg P \wedge R) \vee (Q \wedge R)$

Solution:

[Note that each term has only two statements and so we will make it three]

Now, $(P \wedge Q) \vee (\neg P \wedge R) \vee (Q \wedge R)$

$$\Leftrightarrow (P \wedge Q \wedge T) \vee (\neg P \wedge R \wedge T) \vee (Q \wedge R \wedge T)$$

$$\Leftrightarrow (P \wedge Q \wedge (R \vee \neg R)) \vee (\neg P \wedge R \wedge (Q \vee \neg Q)) \vee (Q \wedge R \wedge (P \vee \neg P))$$

$$\Leftrightarrow (P \wedge Q \wedge R) \vee (P \wedge Q \wedge \neg R) \vee (\neg P \wedge R \wedge Q) \vee (\neg P \wedge R \wedge \neg Q)$$

$$\vee (Q \wedge R \wedge P) \vee (Q \wedge R \wedge \neg P)$$

$$\Leftrightarrow (P \wedge Q \wedge R) \vee (P \wedge Q \wedge \neg R) \vee (\neg P \wedge R \wedge Q) \vee (\neg P \wedge \neg R \wedge Q)$$

omitting identical terms

This is the PDNF

^{to write}
Principal Conjunctive Normal Form: (PCNF)

* [If a formula is equivalent another formula having conjunctions of MAXTERMS only. It is known as the "principal Conjunctive Normal Form" (or) "The Product of Sums Canonical Form".]

It is the dual of the sum of products type. Just as we defined a MINTERM, here we defined a MAXTERM.

If P and Q are two formulas then $P \vee Q$, $P \vee \neg Q$,

$\neg P \vee Q$, and $\neg P \vee \neg Q$ are called the MAXTERMS.

The truth table for these terms will be

P	Q	$P \vee Q$	$P \vee \neg Q$	$\neg P \vee Q$	$\neg P \vee \neg Q$
T	T	T	T	F	F
T	F	T	T	F	T
F	T	T	F	T	F
F	F	F	T	T	T

Note that each of the max term has the truth value F in exactly one place only.

Any formula which is written as the conjunction of these maxterms only is called its principal Conjunctive Normal Form (PCNF) or Product of Sums normal form.

Note:

The rules for writing the PCNF are the same as in PDNF, because the maxterms are the duals of the minterms.

Ex:

1. Find the PCNF of $\neg(P \vee Q)$

Sol: $\neg(P \vee Q)$

$$\Leftrightarrow \neg P \wedge \neg Q$$

$$\Leftrightarrow (\neg P \vee F) \wedge (\neg Q \vee F) \quad \text{Introduce a missing term}$$

$$\Leftrightarrow (\neg P \vee (Q \wedge \neg Q)) \wedge (\neg Q \vee (P \wedge \neg P)) \quad P \wedge \neg P \Leftrightarrow F$$

$$\Leftrightarrow (\neg P \vee Q) \wedge (\neg P \vee \neg Q) \wedge (\neg Q \vee P) \wedge (\neg Q \vee \neg P)$$

$$\Leftrightarrow (\neg P \vee Q) \wedge (\neg P \vee \neg Q) \wedge (\neg Q \vee P) \quad \text{Omitting like terms.}$$

(ii) $\neg(P \rightarrow Q)$

$$\Leftrightarrow \neg(\neg P \vee Q)$$

$$\Leftrightarrow (P \wedge \neg Q)$$

$$\Leftrightarrow (P \vee F) \wedge (\neg Q \vee F)$$

$$\Leftrightarrow (P \vee (Q \wedge \neg Q)) \wedge (\neg Q \vee (P \wedge \neg P))$$

$$\Leftrightarrow (P \vee Q) \wedge (P \vee \neg Q) \wedge (\neg Q \vee P) \wedge (\neg Q \vee \neg P)$$

$$\Leftrightarrow (P \vee Q) \wedge (P \vee \neg Q) \wedge (\neg P \vee \neg Q) \quad \text{Omitting like terms and comm. law.}$$

Rule:

To write PDNF

Write $P \leftrightarrow PAT$ and change T to $(Q \vee \neg Q)$

To write PCNF

Write $P \leftrightarrow PVF$ and change F to $(Q \wedge \neg Q)$.

Write the PCNF of the following:

①. $(\neg P \rightarrow R) \wedge (Q \leftrightarrow P)$

Sol:

$$(\neg P \rightarrow R) \wedge (Q \leftrightarrow P)$$

$$\Leftrightarrow (P \vee R) \wedge ((Q \rightarrow P) \wedge (P \rightarrow Q))$$

$$\Leftrightarrow (P \vee R) \wedge (\neg Q \vee P) \wedge (\neg P \vee Q)$$

$$\Leftrightarrow (P \vee R \vee F) \wedge (P \vee \neg Q \vee F) \wedge (F \vee P \vee Q \vee F)$$

on Introducing missing terms

$$\Leftrightarrow (P \vee R \vee (Q \wedge \neg Q)) \wedge (P \vee \neg Q \vee (R \wedge \neg R)) \wedge (\neg P \vee Q \vee (R \wedge \neg R))$$

Apply distributive law.

$$\Leftrightarrow (P \vee R \vee Q) \wedge (P \vee R \vee \neg Q) \wedge (P \vee \neg Q \vee R) \wedge (P \vee \neg Q \vee \neg R)$$

$$\wedge (\neg P \vee Q \vee R) \wedge (\neg P \vee Q \vee \neg R)$$

Dropping the like terms and using commutative law.

$$\Leftrightarrow (P \vee R \vee Q) \wedge (P \vee \neg Q \vee R) \wedge (P \vee \neg Q \vee \neg R) \wedge (\neg P \vee Q \vee R) \wedge (\neg P \vee Q \vee \neg R)$$

$\equiv$

2. $\neg(P \leftrightarrow Q)$

$$\Leftrightarrow \neg((P \rightarrow Q) \wedge (Q \rightarrow P))$$

$$\Leftrightarrow \neg((\neg P \vee Q) \wedge (\neg Q \vee P))$$

$$\Leftrightarrow \neg(\neg P \vee Q) \vee \neg(\neg Q \vee P)$$

$$\Leftrightarrow (P \wedge \neg Q) \vee (Q \wedge \neg P)$$

$$\Leftrightarrow (P \wedge \neg Q) \vee (Q \wedge \neg P)$$

$$\Leftrightarrow (P \vee Q) \wedge (\neg Q \vee Q) \wedge (P \vee \neg P) \wedge (\neg Q \vee P)$$

$$\Leftrightarrow (P \vee Q) \wedge (\neg P \vee \neg Q) \quad \text{Dropping } P \vee \neg P \text{ and } Q \vee \neg Q$$

$\equiv$

$$3. (P \wedge Q) \vee (\neg P \wedge Q \wedge R)$$

Sol:

$$\begin{aligned}
 & (P \wedge Q) \vee (\neg P \wedge Q \wedge R) \\
 \Leftrightarrow & ((P \wedge Q) \vee \neg P) \wedge ((P \wedge Q) \vee Q) \wedge ((P \wedge Q) \vee R) \\
 \Leftrightarrow & (P \vee \neg P) \wedge (Q \vee \neg P) \wedge (P \vee Q) \wedge (Q \vee Q) \wedge (P \vee R) \wedge (Q \vee R) \\
 \Leftrightarrow & (\neg P \vee Q) \wedge (P \vee Q) \wedge (P \vee R) \wedge (Q \vee R) \quad \text{Drop } P \vee \neg P, Q \vee Q \\
 \Leftrightarrow & (\neg P \vee Q \vee F) \wedge (P \vee Q \vee R) \wedge (P \vee R \vee F) \wedge (Q \vee R \vee F) \quad \text{Introduce missing term} \\
 \Leftrightarrow & (\neg P \vee Q \vee (R \wedge \neg R)) \wedge (P \vee Q \vee (R \wedge \neg R)) \wedge (P \vee R \vee (Q \wedge \neg Q)) \\
 & \quad \wedge (Q \vee R \vee (P \wedge \neg P)) \\
 \Leftrightarrow & (\neg P \vee Q \vee R) \wedge (\neg P \vee Q \vee \neg R) \wedge (P \vee Q \vee R) \wedge (P \vee Q \vee \neg R) \\
 & \quad \wedge (P \vee R \vee Q) \wedge (P \vee R \vee \neg Q) \wedge (Q \vee R \vee P) \wedge (Q \vee R \vee \neg P) \\
 \Leftrightarrow & (\neg P \vee Q \vee R) \wedge (\neg P \vee Q \vee \neg R) \wedge (P \vee Q \vee R) \wedge (P \vee Q \vee \neg R) \\
 & \quad \wedge (P \vee \neg Q \vee R) \wedge (P \vee \neg Q \vee \neg R) \quad \text{omitting like terms}
 \end{aligned}$$

$$4. Q \wedge (P \vee \neg R)$$

Sol:

$$\begin{aligned}
 & Q \wedge (P \vee \neg R) \\
 \Leftrightarrow & (Q \vee F) \wedge (P \vee \neg R) \quad \text{Introduce a missing term} \\
 \Leftrightarrow & Q \vee (P \wedge \neg P) \wedge (P \vee \neg R) \\
 \Leftrightarrow & (Q \vee P) \wedge (Q \vee \neg P) \wedge (P \vee \neg R) \\
 \Leftrightarrow & (P \vee Q) \wedge (\neg P \vee Q) \wedge (P \vee \neg R) \\
 & \quad \underline{\underline{((Q \vee P) \wedge (P \vee \neg R))}}
 \end{aligned}$$

Find the PDNF and PCNF for the following formulas

(i) $(\neg P \vee \neg Q) \rightarrow (P \rightarrow Q)$

(ii) $Q \wedge (P \vee \neg R)$

(iii) $P \rightarrow (P \wedge (Q \rightarrow P))$

(iv) $(Q \rightarrow P) \wedge (\neg P \wedge Q)$

(17)

The theory of inference for statement calculus: (Rules of Reasoning)

Premise:

It is an assumption (or an axiom or a hypothesis).

Def:1 (Tautological implication)

Let A and B be two statement formulas we say that "B logically follows from A" (or) "B is a valid conclusion of the premise A" iff $A \rightarrow B$ is a

Tautology. This we write as $A \Rightarrow B$.

Def:2

From a set of premise $\{H_1, H_2, \dots, H_m\}$ we say that conclusion C follows logically iff

$$\{H_1 \wedge H_2 \wedge \dots \wedge H_m\} \Rightarrow C$$

i.e. $\{H_1 \wedge H_2 \wedge \dots \wedge H_m\} \rightarrow C$ is a tautology.

Meaning:

If we know that H_1 is true and H_2 is true ... and H_m is true, then we can conclude that C is true.

This can be verified by using the truth table.

But this method will be difficult. So we use a different method, which uses two rules R and T.

These two rules are called the rules of inference.

Rule-P:

(A premise) may be introduced at any point in the derivation.

Rule-T:

A formula's may be introduced in a derivation if it is tautologically implied by any one or more all the formulas given earlier in the derivation.

Ex 11

P.T R is a valid inference from the premise

$P \rightarrow Q, Q \rightarrow R$ and P .

	Reasoning
1) $P \rightarrow Q$	Rule P
2) P	Rule P
3) Q	Rule T, (1), (2)
4) $Q \rightarrow R$	Rule P
5) R	Rule T, (3), (4)

We know given below a table of implications which are used in the theory of inference.

Table of Implications:

$$I_1 \quad P \wedge Q \Rightarrow P$$

$$I_2 \quad P \wedge Q \Rightarrow Q$$

$$I_3 \quad P \Rightarrow P \vee Q$$

$$I_4 \quad Q \Rightarrow P \vee Q$$

$$I_5 \quad \neg P \Rightarrow P \rightarrow Q$$

$$I_6 \quad Q \Rightarrow P \rightarrow Q$$

$$I_7 \quad \neg(P \rightarrow Q) \Rightarrow P$$

$$I_8 \quad \neg(P \rightarrow Q) \Rightarrow \neg Q$$

$$I_9 \quad P, Q \Rightarrow P \wedge Q$$

$$I_{10} \quad \neg P, P \vee Q \Rightarrow Q \text{ (Disjunctive Syllogism)}$$

$$I_{11} \quad P, P \rightarrow Q \Rightarrow Q \text{ (modus Ponens)}$$

$$I_{12} \quad \neg Q, P \rightarrow Q \Rightarrow \neg P \text{ (modus Tollens)}$$

$$I_{13} \quad P \rightarrow Q, Q \rightarrow R \Rightarrow P \rightarrow R \text{ (hypothetical syllogism)}$$

$$I_{14} \quad P \vee Q, P \rightarrow R, Q \rightarrow R \Rightarrow R \text{ (dilemma)}$$

Table of equivalent

E_1	$\neg\neg P \Leftrightarrow P$	
E_2	$P \wedge Q \Leftrightarrow Q \wedge P$	Commutative laws
E_3	$P \vee Q \Leftrightarrow Q \vee P$	
E_4	$(P \wedge Q) \wedge R \Leftrightarrow P \wedge (Q \wedge R)$	Associative laws
E_5	$(P \vee Q) \vee R \Leftrightarrow P \vee (Q \vee R)$	
E_6	$P \wedge (Q \vee R) \Leftrightarrow (P \wedge Q) \vee (P \wedge R)$	Distributive laws
E_7	$P \vee (Q \wedge R) \Leftrightarrow (P \vee Q) \wedge (P \vee R)$	
E_8	$\neg(P \wedge Q) \Leftrightarrow \neg P \vee \neg Q$	De Morgan's laws
E_9	$\neg(P \vee Q) \Leftrightarrow \neg P \wedge \neg Q$	
E_{10}	$P \vee P \Leftrightarrow P$	
E_{11}	$P \wedge P \Leftrightarrow P$	
E_{12}	$R \vee (P \wedge T) \Leftrightarrow R$	
E_{13}	$R \wedge (P \vee T) \Leftrightarrow R$	
E_{14}	$P \rightarrow Q \Leftrightarrow \neg P \vee Q$	
E_{15}	$\neg(P \rightarrow Q) \Leftrightarrow P \wedge \neg Q$	
E_{16}	$P \rightarrow Q \Leftrightarrow \neg Q \rightarrow \neg P$	
E_{17}	$P \rightarrow (Q \rightarrow R) \Leftrightarrow (P \wedge Q) \rightarrow R$	

Ex: 2

S.T RVS, follows logically from the premises
 $C \vee D, C \vee D \rightarrow \neg H, \neg H \rightarrow (A \wedge \neg B)$ and $(A \wedge \neg B) \rightarrow (R \vee S)$

Solution:

We use the theory of inference to derive the conclusion.

- $\{1\}$ 1) $C \vee D \rightarrow 7H$ P
 $\{2\}$ 2) $7H \rightarrow (A \wedge 7B)$ P
 $\{1, 2\}$ 3) $C \vee D \rightarrow (A \wedge 7B)$ $T, (1), (2)$ and I_{13}
 $\{4\}$ 4) $(A \wedge 7B) \rightarrow (R \vee S)$ P
 $\{1, 2, 4\}$ 5) $(C \vee D) \rightarrow (R \vee S)$ $T, (3), (4)$ and I_{13}
 $\{6\}$ 6) $C \vee D$ P
 $\{1, 2, 4, 6\}$ 7) $R \vee S$ $T, (5), (6)$ and I_{11}

Ex: 3

Show that $S \vee R$ is tautologically implied by

$$(P \vee Q) \wedge (P \rightarrow R) \wedge (Q \rightarrow S)$$

Solution:

- $\{1\}$ 1) $P \vee Q$ P
 $\{1\}$ 2) $7P \rightarrow Q$ $T, (1), E_1$ and E_{14}
 $\{3\}$ 3) $Q \rightarrow S$ P
 $\{1, 3\}$ 4) $7P \rightarrow S$ $T, (2), (3)$ and I_{13}
 $\{1, 3\}$ 5) $7S \rightarrow P$ $T, (4)$ and E_{13}
 $\{6\}$ 6) $P \rightarrow R$ P
 $\{1, 3, 6\}$ 7) $7S \rightarrow R$ $T, (5), (6)$ and I_{13}
 $\{1, 3, 6\}$ 8) $S \vee R$ $T, (7), E_1$ and E_{14}

Ex: 4

Show that $7Q, P \rightarrow Q \Rightarrow 7P$

Solution:

- $\{1\}$ 1) $P \rightarrow Q$ P
 $\{1\}$ 2) $7Q \rightarrow 7P$ $T, (1), E_{14}$
 $\{3\}$ 3) $7Q$ P
 $\{1, 3\}$ 4) $7P$ $T, (2), (3)$ and I_{11}

Ex: 5

S.T $R \wedge (P \vee Q)$ is a valid conclusion from the premises $P \vee Q$, $Q \rightarrow R$, $P \rightarrow M$ and $\neg M$

Solution:

{1} 1) $P \rightarrow M$ 4) P

{2} 2) $\neg M$ 5) P

{1, 2} 3) $\neg P$ 7) $\neg$, (1), (2) and I_{12}

{4} 4) $P \vee Q$ 5) P

{1, 2, 4} 5) Q 6) $\neg$, (1), (4) and I_{10}

{6} 6) $Q \rightarrow R$ 7) P

{1, 2, 4, 6} 7) R 8) $\neg$, (5), (6) and I_{11}

{1, 2, 4, 6} 8) $R \wedge (P \vee Q)$ 9) $\neg$, (4), (7) and I_9

Rule CP: [C - Conclusion]
[P - Premises]

It is the third rule of inference. It is usually applied in problems where the conclusion is of the form $A \rightarrow B$. The rule says that along with the premise P , take 'A' also as a premise and prove 'B'.

Ex: 1 Show that $R \rightarrow S$ can be derived from the premises $P \rightarrow (Q \rightarrow S)$, $\neg R \vee P$ and Q .

Solution:

Here, there are 3 premises given.

Include R also as a premise and derive S .

(1) $\neg R \vee P$

P

(2) R

P(Conditional premise)

(3) P

T, (1), (2) and I_{10}

(4) $P \rightarrow (Q \rightarrow S)$

P

(5) $Q \rightarrow S$

T, (3), (4) and I_{11}

(6) Q

P

(7) S

T, (5), (6) and I_{11}

(8) $R \rightarrow S$

CP

We now give examples of derivation using statements in English:

Ex: 1

If there was a ball game, then travelling was difficult. If they arrived on time, then travelling was not difficult. They arrived on time. Therefore there was no ball game.

Solution:

Let P: There was a ball game

Q: Travelling was difficult

R: They arrived on time

The premises are,

$P \rightarrow Q, R \rightarrow \neg Q, R \Rightarrow \neg P$

{1}	(1)	$R \rightarrow \neg Q$	P
{1,2}	(2)	R	P
{1,2}	(3)	$\neg Q$	T, (1), (2) $\& I_{11}$
{1,3}	(4)	$P \rightarrow Q$	P
{1,2,4}	(5)	$\neg P$	T, (3), (4) $\& I_{12}$

- ② " If A works hard, then either B or C will enjoy themselves. If B enjoys himself; then A will not work hard. If D enjoys himself, then C will not. Therefore if A work hard, then D will not enjoy himself.

Solution:

Let A : A work hard

B : B enjoys himself

C : C enjoys himself

D : D enjoys himself

~~Then we get~~

The premises are,

$$A \rightarrow (B \vee C); B \rightarrow \neg A, D \rightarrow \neg C \Rightarrow A \rightarrow \neg D$$

Inconsistency of Premises:

To show that, a set of premises

$\{H_1, H_2, \dots, H_m\}$ is inconsistent.

We show that $(H_1 \wedge H_2 \wedge \dots \wedge H_m) \Rightarrow$ a contradiction

Ex 1.1

Show that the following premises are inconsistent

(i) If Jack misses many classes through illness, then he fails in high school.

(ii) If Jack fails in high school, then he is uneducated.

(iii) If Jack reads a lot of books, then he is not uneducated.

(iv) Jack misses many classes through illness and reads a lot of books.

Solution:

There are four premises and a conclusion

Let E : Jack misses many classes through illness

S : He fails in high school

A : He is uneducated

H : He reads a lot of books.

Then the given premises are

$$1. E \rightarrow S, S \rightarrow A, H \rightarrow \neg A, E \wedge H$$

$$(1) E \rightarrow S \quad P$$

$$(2) S \rightarrow A \quad P$$

$$(3) E \rightarrow A \quad T, (1), (2) \& I_{13}$$

$$(4) H \rightarrow \neg A \quad P$$

$$(5) A \rightarrow \neg H \quad T, (4) \text{ and } E_{18} [P \rightarrow Q \Leftrightarrow \neg Q \rightarrow \neg P]$$

$$(6) E \rightarrow \neg H \quad T, (3), (5) \text{ and } I_{13}$$

$$(7) E \vee \neg H \quad T, (6) \text{ and } E_{16}$$

$$(8) \neg(E \wedge H) \quad T, (7) \text{ and } E_8 : (E \vee \neg H) \leftarrow A$$

The final step is a contradiction because we must get only $E \wedge H$.

$\therefore$ The given set of premises is inconsistent.

We can also prove the inconsistency by

another method.

Suppose we want to prove the conclusion from a set of premises.

We assume that C is false.

Then $\neg C$ is true.

We take $\neg C$ also as a premise and P.T.

The new set of premises is inconsistent.

This will prove that C must be true.

Q. S.T. $\neg(P \wedge Q)$ logically follows from $\neg P \wedge \neg Q$.

Sol:

Here the conclusion is $\neg(P \wedge Q)$.

So we take $\neg(P \wedge Q)$ as a premise.

(21)

So we have two premises $\neg\neg(P \wedge Q)$ and $\neg P \wedge \neg Q$

We prove that these are inconsistent.

(1) $\neg\neg(P \wedge Q)$ New premise

(2) $P \wedge Q$ $\neg, (1)$ and E_1

(3) P $\neg, (2)$ and I_1

(4) $\neg P \wedge \neg Q$ P

(5) $\neg P$ $\neg, (4)$ and I_1

(6) $P \wedge \neg P$ $\neg, (3), (5)$ and I_3

This is a contradiction

hence $\therefore$ The two premises are inconsistent

$\therefore \neg\neg(P \wedge Q)$ is False

$\therefore \neg(P \wedge Q)$ is True.

③ S.T the following premises are inconsistent

$P \rightarrow Q, P \rightarrow R, Q \rightarrow \neg R, P$

Solution:

(1) $P \rightarrow Q$ P

(2) $Q \rightarrow \neg R$ P

(3) $P \rightarrow \neg R$ $\neg, (1), (2)$ and I_3

(4) $P \rightarrow R$ P

(5) $R \rightarrow \neg P$ $\neg, (3)$ and I_3

(6) $P \rightarrow \neg P$ $\neg, (4), (5)$ and I_{13}

(7) P P

(8) $\neg P$ $\neg, (6), (7)$ and I_1

Which is a contradiction

$\therefore$ The set of premises is inconsistent.

The theory of inference is for Predicate calculus:

Consider the statement, Ram is a boy,

Here, 'Ram' is a name, denoted by 'r'.

"is a boy" is the predicate denoted by 'B'.

So "Ram is a boy" is written as $B(r)$.

Thus names are written in small letters and predicates are written in Capital letters.

In general any statement of the type,

"p is q" is written as $Q(p)$.

A Predicate requiring 'm' names is called an m place predicate.

So here $B(r)$ is a one place predicate.

Consider the statement,

"Rama is taller than Krishna"

Here, there are two names and the predicate is "is taller than", if we denote this predicate by T.

We can denote the sentence as $T(r, k)$

Note that, $T(k, r) \neq T(r, k)$

This is a 2 place predicate and the order is important.

Compound Statement:

We can use the connectives $\wedge, \vee, \neg, \rightarrow, \leftrightarrow$ etc to form a compound statement.

For Ex:

Here $B(x) \wedge T(x, k)$ means that

"Ram is a boy and Ram is taller than Krishna".

iii) If $R(a)$: This apple is Red

and $B(p)$: Painting is beautiful

Then $R(a) \rightarrow B(p)$ is valid

iii) $\neg R(a)$, $R(a) \Leftrightarrow B(p)$, etc are all valid.

In general if $a_1, a_2, \dots, a_n$ are n names and B is an n place predicate we can write as

$B(a_1, a_2, \dots, a_n)$

Statement function:

A statement function of one variable is defined as an expression having a predicate symbol and a variable.

For ex:

If M : is a mortal, is a predicate

Ram is mortal.

Krishna is mortal and

Govinda is mortal

can be written as $M(r)$, $M(k)$, $M(g)$.

In general if we write $M(x)$, then it means that " x is mortal" so $M(x)$ is a function, and $M(r)$, $M(k)$ and $M(g)$ are called substitution instances of $M(x)$.

Quantifier:

(i) "For every x (or) for any x " is denoted by $(\forall x)$ (or) by (x) . It is called "Universal Quantifier".

(2) "There is atleast one x such that", "there exists some x such that", (or) "for some x ", is denoted by $(\exists x)$. It is called Existential Quantifier.

Note:

" $(\forall)$ " (or) " $(\exists)$ " are called Quantification Symbols.

Ex:

Consider the following statements.

$M(x)$: x is a man

$H(x)$: x is a mortal

$A(x)$: x is an apple

$R(x)$: x is Red.

We give below sentences with meanings

(1) $(\forall x) M(x)$ is a statement which is read as,

for all x , x is a man

(2) For all x , if x is a man, then he is mortal

Written as $(\forall x) (M(x) \rightarrow H(x))$

(3) For all x , if x is an apple, then x is red,

write it as $(\forall x) (A(x) \rightarrow R(x))$

11/9

Consider the statements

$M(x)$: x is a man

$C(x)$: x is clever

then we can write

(1) "there exists a man" as $(\exists x) M(x)$.

(2) "Some men are clever" as $(\exists x) (M(x) \rightarrow C(x))$

Free and Bound variables:

If a formula contains a part like $(\forall x) P(x)$ or $(\exists x) P(x)$, such a part is called an x -bound part. x is called bound occurrence.

If x is not bound it is called a free occurrence.

For ex:

$(x) (P(x) \rightarrow Q(x))$, $(x) (P(x) \rightarrow (\exists y) Q(x))$ are all bound occurrences

But in $(x) P(x) \rightarrow Q(x)$

$P(x)$ is an x bound part and $Q(x)$ is a free occurrence.

Inference theory of predicate calculus:

It is similar to that of statement calculus, except that statements are replaced by predicates.

So the rules are similar here also.

For ex:

If P and Q are taken as $(x) R(x)$ and $(\exists x) S(x)$,
 $P \vee Q$ is the same as $(x) R(x) \vee (\exists x) S(x)$.

III) $P \rightarrow Q \Leftrightarrow \neg P \vee Q$ is equivalent to
 $(x) R(x) \rightarrow (\exists x) S(x) \Leftrightarrow \neg (x) R(x) \vee (\exists x) S(x)$

Prime Formula:

A predicate formula is called a Prime (atomic) formula, if it does not contain any connectives.

In such cases, all the implications and equivalences of the statement calculus can be taken true for the Predicate calculus also.

For ex:

If $B(x)$ and $C(x, y)$ are two prime formulae, then $B(x) \wedge C(x, y) \Leftrightarrow C(x, y) \wedge B(x)$
 $\neg \neg B(x) \Leftrightarrow B(x)$.

Apart from this, we give below some important formulas used in Predicate calculus.

Equivalences:

$$\left. \begin{array}{l} (1) \neg(x) A(x) \Leftrightarrow (\exists x) \neg A(x) \\ (2) \neg(\exists x) A(x) \Leftrightarrow (x) \neg A(x) \end{array} \right\} \textcircled{X}$$

Rules for Derivation of Inference:

The method of derivation of inference using Predicate formulas is similar to that of statement calculus using rules P and T and also using certain other rules.

1. Rule P:-

Introducing a premise at any time of derivations.

2. Rule T:

Introducing any formula which follows logically from the previous steps of the derivation.

3. Rule US: (Universal Specification)

We can replace the general function $(x) A(x)$ by a specific value $A(y)$.

4. Rule ES: (Existential Specifier)

We can replace the general function $(\exists x) A(x)$ by a specific value $A(y)$.

i, Using $\forall$ and $\exists$, we eliminate the quantifiers and the proof is given as in the case of Statement calculus. After the proof is over, we go back to the general value from the particular value.

Rule $\forall$ G: (Universal Generalisation)

$A(y)$ is replaced by $(\forall x) A(x)$.

Rule $\exists$ G: (Existential Generalisation)

$A(y)$ is replaced by $(\exists x) A(x)$.

Example:

1. S.T. $(\forall x) (H(x) \rightarrow M(x)) \wedge H(s) \Rightarrow M(s)$

Hint: [There are two premises $(x) (H(x) \rightarrow M(x))$ and $H(s)$. We have to P.T. $M(s)$ follows logically from these two premises].

Sol:

- | | | |
|-----|---------------------------------------|----------------------------------|
| (1) | $(\forall x) (H(x) \rightarrow M(x))$ | P |
| (2) | $H(s) \rightarrow M(s)$ | (1), $\forall$ S |
| (3) | $H(s)$ | P |
| (4) | $M(s)$ | T, (2), (3) and $\rightarrow$ I. |

② P.T. $(\forall x) (P(x) \rightarrow Q(x)) \wedge (\forall x) (Q(x) \rightarrow R(x)) \Rightarrow (\forall x) (P(x) \rightarrow R(x))$

Sol:

- | | | |
|-----|---------------------------------------|----------------------------------|
| (1) | $(\forall x) (P(x) \rightarrow Q(x))$ | P |
| (2) | $P(y) \rightarrow Q(y)$ | (1), $\forall$ S |
| (3) | $(\forall x) (Q(x) \rightarrow R(x))$ | P |
| (4) | $Q(y) \rightarrow R(y)$ | (3), $\forall$ S |
| (5) | $P(y) \rightarrow R(y)$ | T, (2), (4) and $\rightarrow$ I. |
| (6) | $(\forall x) (P(x) \rightarrow R(x))$ | (5) $\forall$ G. |

③ S.T. $(\exists x) M(x)$ follows logically from the premises $(\forall x) (H(x) \rightarrow M(x))$ and $(\exists x) H(x)$.

Sol:

- (1) $(\exists x) H(x)$ P
- (2) $H(y)$ (1), ES y fixed
- (3) $(\exists x) (H(x) \rightarrow m(x))$ P
- (4) $H(y) \rightarrow m(y)$ (3), US
- (5) $m(y)$ T, (2), (4) and I_{11}
- (6) $(\exists x) m(x)$ (5), EG.

Note:

In step (6) we cannot write the quantifier (x) because y is fixed in step (2).

④ P.T $(\exists x) (P(x) \wedge Q(x)) \Rightarrow (\exists x) P(x) \wedge (\exists x) Q(x)$.

Sol:

- (1) $(\exists x) (P(x) \wedge Q(x))$ P
- (2) $P(y) \wedge Q(y)$ (1), ES
- (3) $P(y)$ T, (2), and I_1 $[P \wedge Q \rightarrow P]$
- (4) $Q(y)$ T, (2) and I_2 $[P \wedge Q \rightarrow Q]$
- (5) $(\exists x) P(x)$ (3), EG
- (6) $(\exists x) Q(x)$ (4), EG
- (7) $(\exists x) P(x) \wedge (\exists x) Q(x)$ T, (5), (6) and I_9 $[P, Q \Rightarrow P \wedge Q]$

⑤ S.T from (a) $(\exists x) (F(x) \wedge S(x) \rightarrow (y)(m(y) \rightarrow w(y)))$

(b) $(\exists y) (m(y) \wedge \neg w(y))$, the conclusion (c) $(F(x) \rightarrow \neg S(x))$ follows.

Sol:

- (1) $(\exists y) (m(y) \wedge \neg w(y))$ P y fixed
- (2) $m(z) \wedge \neg w(z)$ (1), ES
- (3) $\neg (m(z) \rightarrow w(z))$ T, (2), E_{17}
- (4) $(\exists y) \neg (m(y) \rightarrow w(y))$ (3), EG
- (5) $\neg (y)(m(y) \rightarrow w(y))$ (4) and $(\exists x) \neg A(x) \Leftrightarrow \neg (x) A(x)$.
- (6) $(\exists x) (F(x) \wedge S(x) \rightarrow (y)(m(y) \rightarrow w(y)))$ P
- (7) $\neg (\exists x) (F(x) \wedge S(x))$ T, (5), (6) and I_{12}

$$(8) \quad (x) \neg (F(x) \wedge S(x)) \quad T, (7) \text{ and } \neg(\exists x) A(x) \Leftrightarrow (x) \neg A(x)$$

$$(9) \quad \neg (F(a) \wedge S(a)) \quad (8), US$$

$$(10) \quad \neg F(a) \vee \neg S(a) \quad T, (9) \text{ and } E_0$$

$$(11) \quad F(a) \rightarrow \neg S(a) \quad T, (10) \text{ and } E_{16}$$

$$(12) \quad (x)(F(x) \rightarrow \neg S(x)) \quad (11), UG.$$

⑥ Show that $(x)(P(x) \vee Q(x)) \Rightarrow (x)P(x) \vee (\exists x)Q(x)$

Solution:

To show the conclusion $(x)P(x) \vee (\exists x)Q(x)$

to be True, we take it as False.

$\neg((x)P(x) \vee (\exists x)Q(x))$ will be True.

We take this also as a premise.

$$(1) \quad \neg((x)P(x) \vee (\exists x)Q(x)) \quad \text{Assumed premise}$$

$$(2) \quad \neg(x)P(x) \wedge \neg(\exists x)Q(x) \quad T, (1), E_7$$

$$(3) \quad \neg(x)P(x) \quad T, (2), I_1$$

$$(4) \quad (\exists x) \neg P(x) \quad (3) \text{ and } \neg(x)A(x) \Leftrightarrow (x) \neg A(x)$$

$$(5) \quad \neg(\exists x)Q(x) \quad T, (2), I_2$$

$$(6) \quad (x) \neg Q(x) \quad (5) \text{ and } \neg(\exists x)A(x) \Leftrightarrow (x) \neg A(x)$$

$$(7) \quad \neg P(y) \quad (4), ES$$

$$(8) \quad \neg Q(y) \quad (6), US$$

$$(9) \quad \neg P(y) \wedge \neg Q(y) \quad T, (7), (8) \text{ and } I_9$$

$$(10) \quad \neg(P(y) \vee Q(y)) \quad T, (9) \text{ and } E_9$$

$$(11) \quad (x)(P(x) \vee Q(x)) \quad P$$

$$(12) \quad P(y) \vee Q(y) \quad (11), US$$

$$(13) \quad \neg(P(y) \vee Q(y)) \wedge (P(y) \vee Q(y)) \quad T, (10), (12)$$

a Contradiction

$\therefore (x)P(x) \vee (\exists x)Q(x)$ is True.